



EAST WEST UNIVERSITY

Lab Report-1

Name: Yougshree Saha Urmy

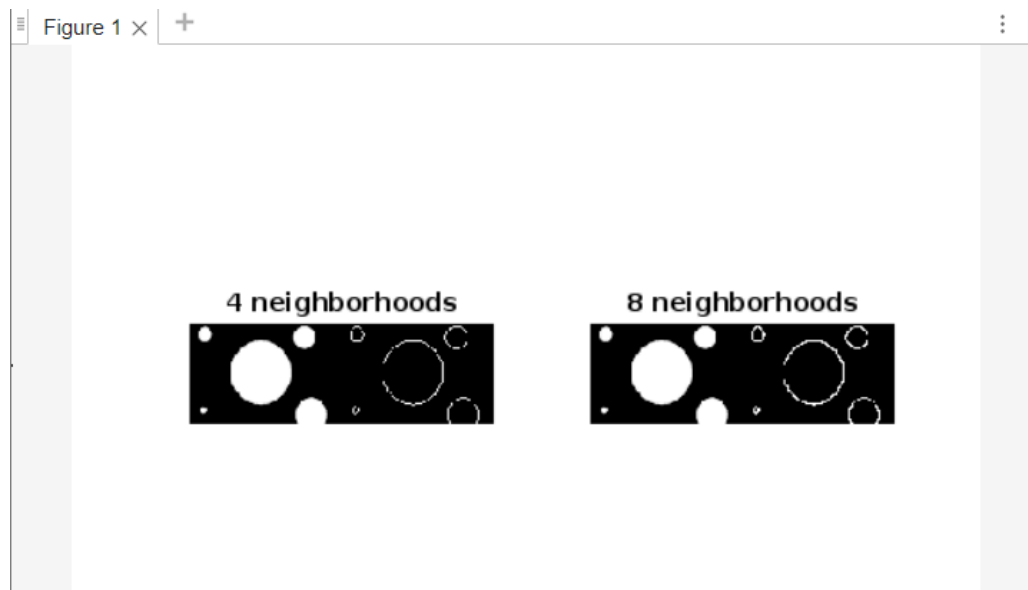
Id: 2021-1-60-127

Section: 01

1. Determine the perimeter of an object by using 4 connected neighborhoods and 8 connected neighborhoods.

Answer:

```
img = imread("img1.png");
imshow(img);
n4=bwperim(img, 4);
n8=bwperim(img, 8);
subplot(1,2,1), imshowpair(img,n4, 'montage');
title("4 neighborhoods");
subplot(1,2,2), imshowpair(img,n8, 'montage');
title("8 neighborhoods");
```



2. Create a binary image using a threshold.

Answer:

```
img = imread("img2.png");
if size(img,3)==3
    img=rgb2gray(img);
end
threshold= 128;
binary_img= img>threshold;
imshowpair(img, binary_img, 'montage');
title("Binary image");
imshow(binary_img);
```

Figure 2 x +



3. Determine the number of objects in the binary image generated in Question 2 using the concept of connectivity.

Answer:

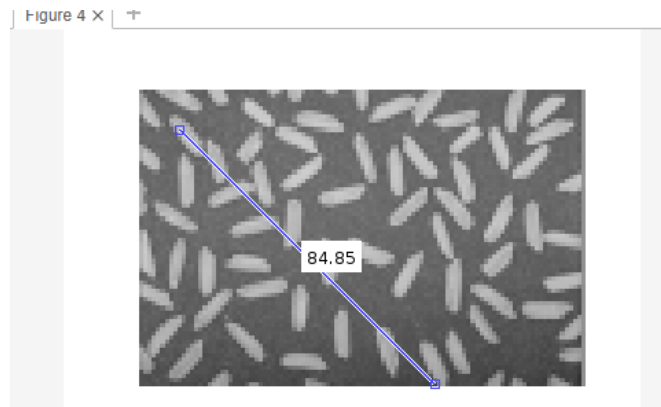
```
img = imread("img2.png");  
if size(img,3)==3  
    img=rgb2gray(img);  
end  
threshold= 128;  
binary_img= img>threshold;  
imshowpair(img, binary_img,'montage');  
title("Binary image");  
imshow(binary_img);  
cc = bwconncomp(binary_img, 8);  
disp(['The number of objects in the image is: ', num2str(cc.NumObjects)]);
```



4. Find the Euclidean distance between two points of the image.

Answer:

```
img = imread ("img2.png");  
imshow(img)  
h= imdistline(gca,[10,70], [10,70]);
```

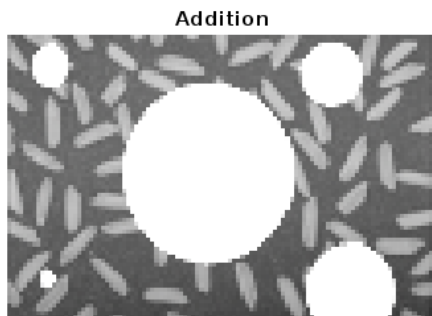


5. Apply the following operations using Fig.1 and Fig.2:

- a. Addition
- b. Subtraction
- c. Multiplication
- d. Division

Answer:

```
img_1 = imread ("img1.png");  
img_2 = imread ("img2.png");  
A = img_1 + img_2  
imshow(A)  
title("Addition")
```



Columns 91 through 105

166	98	98	103	100	96	101	101	99	101	101	101	99	94	144
158	98	100	101	100	96	112	134	132	124	103	99	100	95	144
142	98	103	100	101	142	192	198	198	195	102	101	103	97	145
117	101	103	100	154	198	197	195	195	149	98	104	103	97	145
99	108	99	159	199	196	197	196	143	92	103	102	109	97	144
104	101	151	199	196	197	193	135	98	100	100	101	105	99	145
100	124	195	197	198	189	119	95	95	94	97	99	108	94	144
99	157	190	196	188	113	96	132	165	157	134	136	107	100	145
102	120	183	176	112	132	179	191	182	182	186	192	120	94	145
105	101	106	144	177	194	193	190	187	187	179	126	99	103	145
102	129	180	191	194	196	195	192	183	144	101	102	107	93	145
115	178	189	200	198	190	167	135	98	96	102	103	106	105	145
120	190	195	176	141	115	97	101	103	104	103	104	105	100	144
103	109	104	105	162	190	124	104	102	104	102	108	103	97	145
108	104	110	177	195	190	141	102	99	105	101	106	104	94	145
106	118	183	198	195	187	109	109	106	109	101	106	105	104	144
124	186	197	198	194	139	102	103	102	103	102	106	105	103	144
190	198	197	193	138	101	112	106	102	102	103	109	112	105	145
197	196	193	143	101	107	103	103	108	108	102	106	112	99	144
192	189	141	106	111	105	103	105	110	106	103	98	102	122	147
167	133	103	107	103	108	109	102	107	99	97	135	175	195	148
100	101	100	100	100	100	100	100	100	100	100	100	100	100	140

```
img_1 = imread ("img1.png");
img_2 = imread ("img2.png");
B = img_1 - img_2
imshow(B)
title("Substruction")
```

Substruction

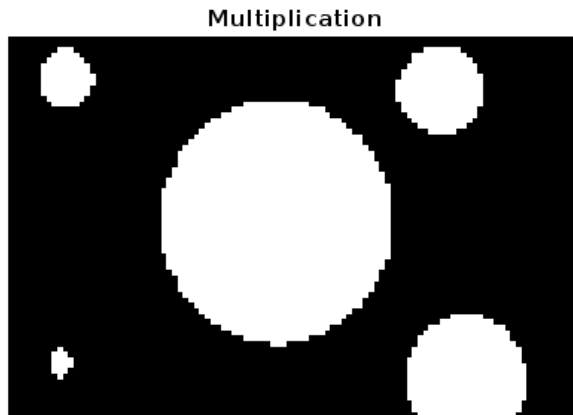


Column 105

0
0
0
0
0
0
0
0
0
0
0

```
img_1 = imread ("img1.png");
img_2 = imread ("img2.png");
```

```
C = immultiply(img_1, img_2)
imshow(C)
title("Multiplication")
```



Column 105

[illegible]

```
img_1 = imread ("img1.png");
img_2 = imread ("img2.png");
D = imdivide(img_1, img_2)
imshow(D)
title("Division")
```



Columns 79 through 105

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0													
2	2	2	2	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0												
2	2	2	3	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0												
2	2	3	2	2	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0												
1	2	3	2	3	2	2	1	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0												
1	2	3	3	2	2	2	3	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0												
2	2	3	2	2	3	3	3	1	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0												
2	2	3	2	2	2	2	3	1	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0												
1	1	2	2	2	1	2	3	2	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0												

6. Apply the following operations using Fig.1 and Fig.2:

a. AND

b. OR

c. NOT

Answer:

```
img_1 = imread ("img1.png");  
img_2 = imread ("img2.png");  
A = and(img_1, img_2)
```

Columns 100 through 105

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

```
img_1 = imread ("img1.png");  
img_2 = imread ("img2.png");  
A = or(img_1, img_2)
```

Columns 100 through 105

[illegible]

```
img_1 = imread ("img1.png");  
img_2 = imread ("img2.png");  
A = not(img_1, img_2)
```

Columns 100 through 105

1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1

7. Adjust the contrast of the following image.

Answer:

```
img = imread("img3.png");  
L= 2^8;  
neg= (L-1)-img;  
imshowpair(img, neg, 'montage')
```



8. Brighten the following image

```

original_image= imread("img4.jpg");
brightened_image= imadjust(original_image, [0 1], [0.2 1]);
figure;
subplot(1, 2, 1),
imshow(original_image),
title('Original Image');
subplot (1, 2, 2),
imshow(brightened_image),
title('Brightened Image');

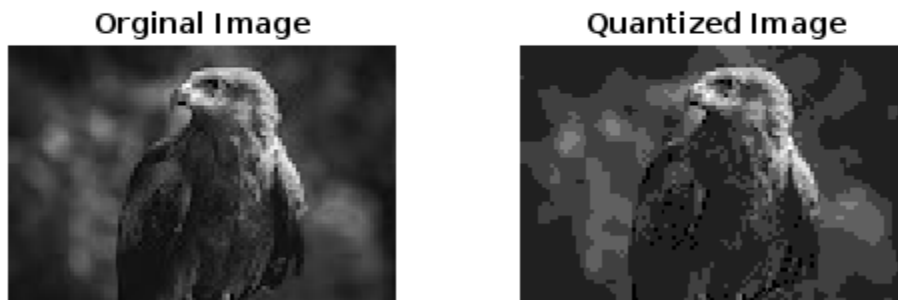
```

9. Quantize the Grayscale image by 8 levels.

```

original_image = imread("img4.jpg");
quantized_image = imdivide(original_image, 32)*32;
figure;
subplot(1,2,1), imshow(original_image),
title('Original Image');
subplot(1,2,2), imshow(quantized_image, []),
title('Quantized Image');

```



10. Find the digital negative of the image.

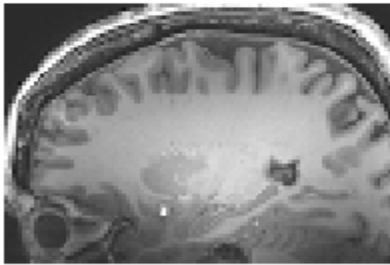
```

img = imread ("img3.png");
subplot(1,2,1), imshow(img);
title("Original image");

```

```
% levels of the 8-bit image
L= 2 ^ 8;
% finding the negative
neg = (L - 1) - img;
subplot(1, 2, 2),
% displaying the negative image
imshow(neg);
title("Negative Image")
```

Original image



Negative Image

